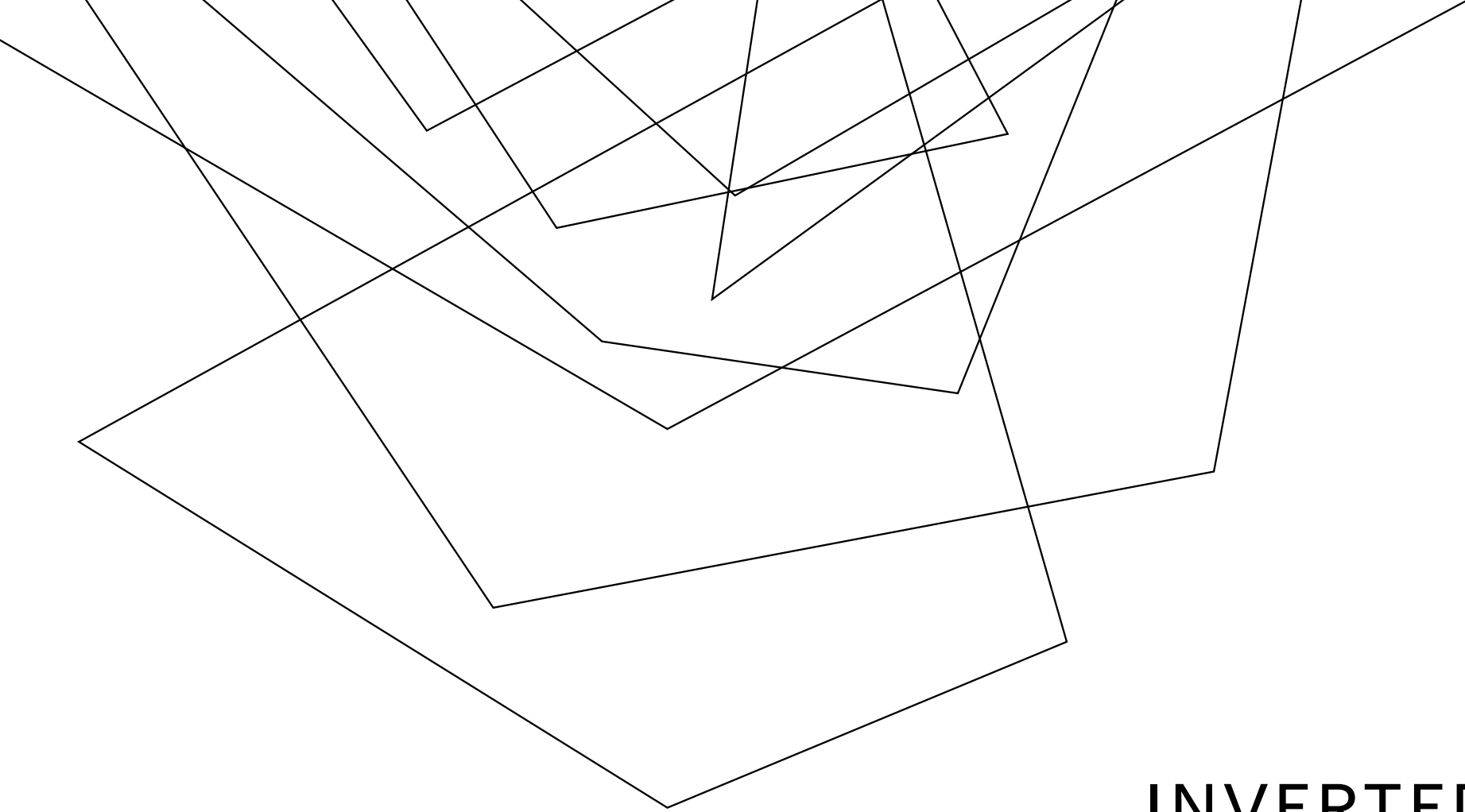




INVERTED PENDULUM

INTERNATIONAL TEAM 5

SETBACKS

Frying the L298N sensor (motor driver)

Frying the motor encoder

Not being able to connect with the AS5600 sensor

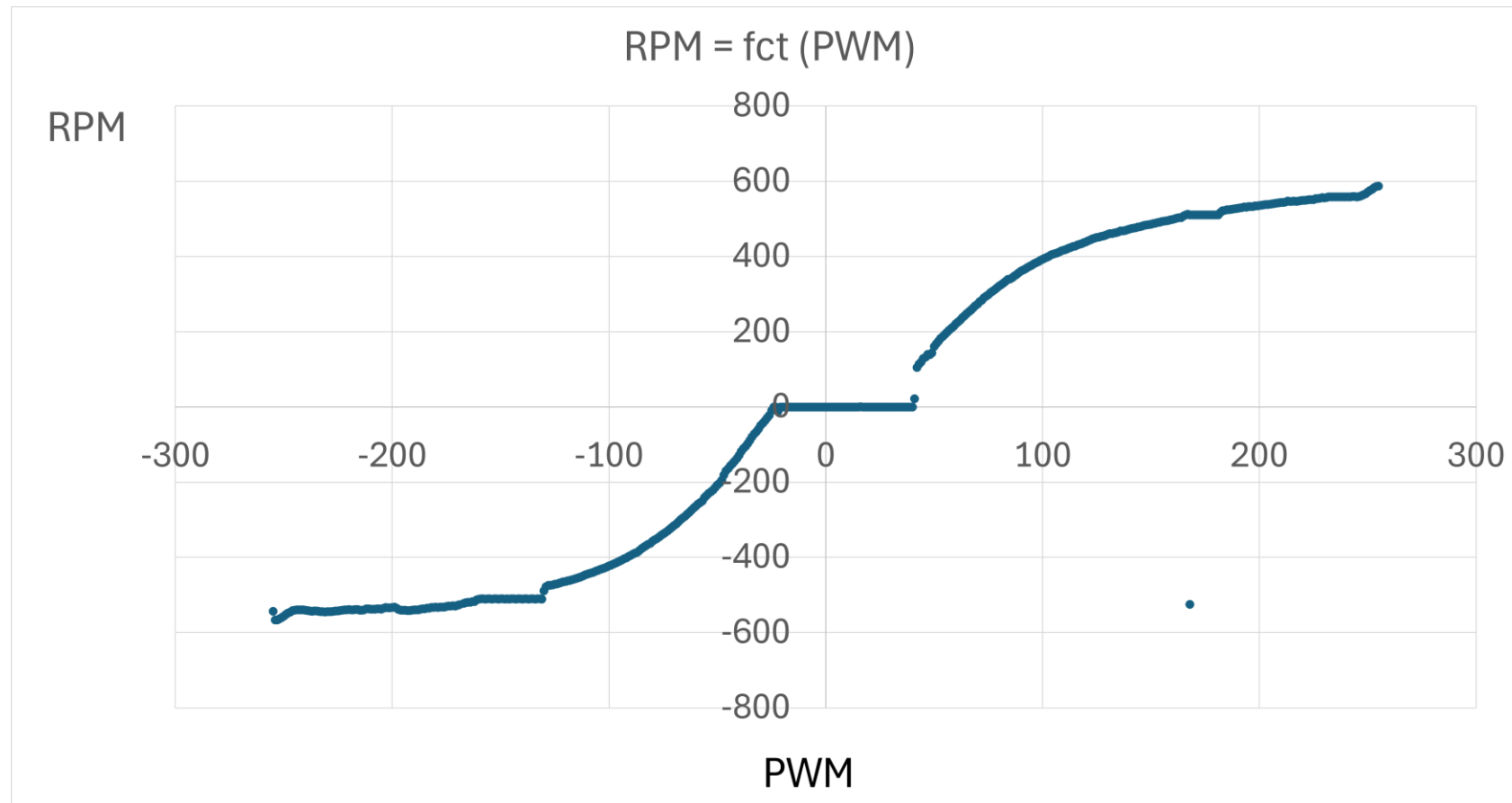
Accelerating wheel

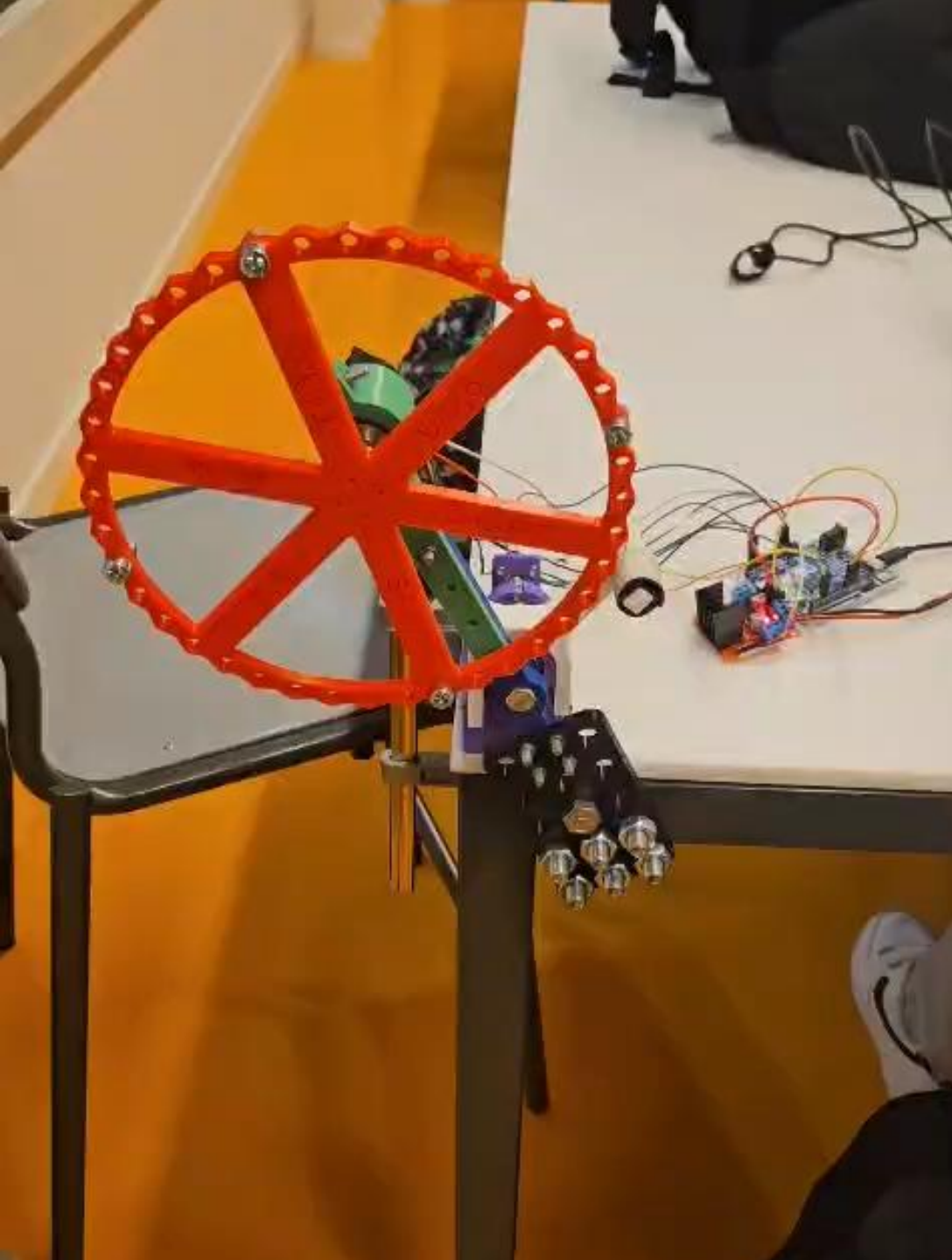
Breaking the 3D printed parts

WORK PERFORMED

- Characterize the motor torque–speed curve
- Implementing a code to print the angle
- Stabilising with the iron bearings
- Stabilize the pendulum with 1 PID (angle)
- Stabilize the pendulum with a second PID (speed)
- Printing the results

CURVES AND GRAPHS





ACCOMPLISHMENTS

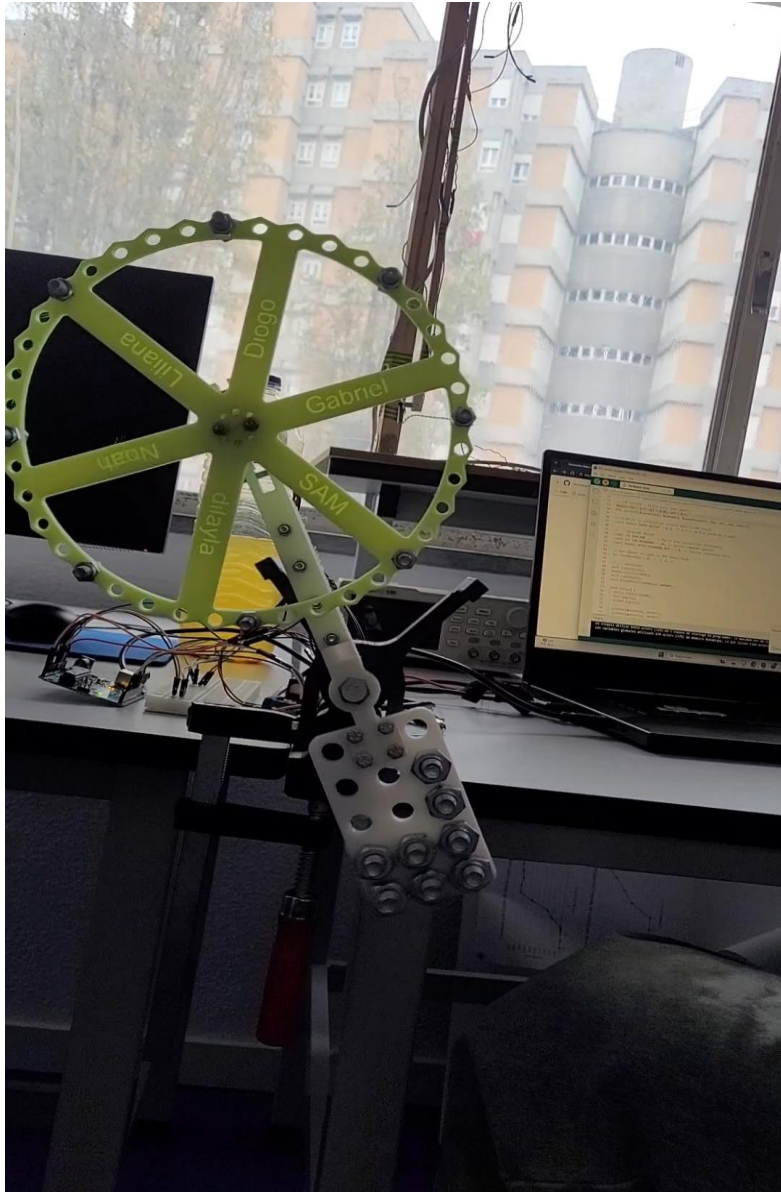
- ❖ Stabilised with the dummy bearings
- ❖ Partially stabilised with the second PID-controller and iron bearings

NEXT OBJECTIVE

Second PID to control the wheel's speed

PID tuning strategy :

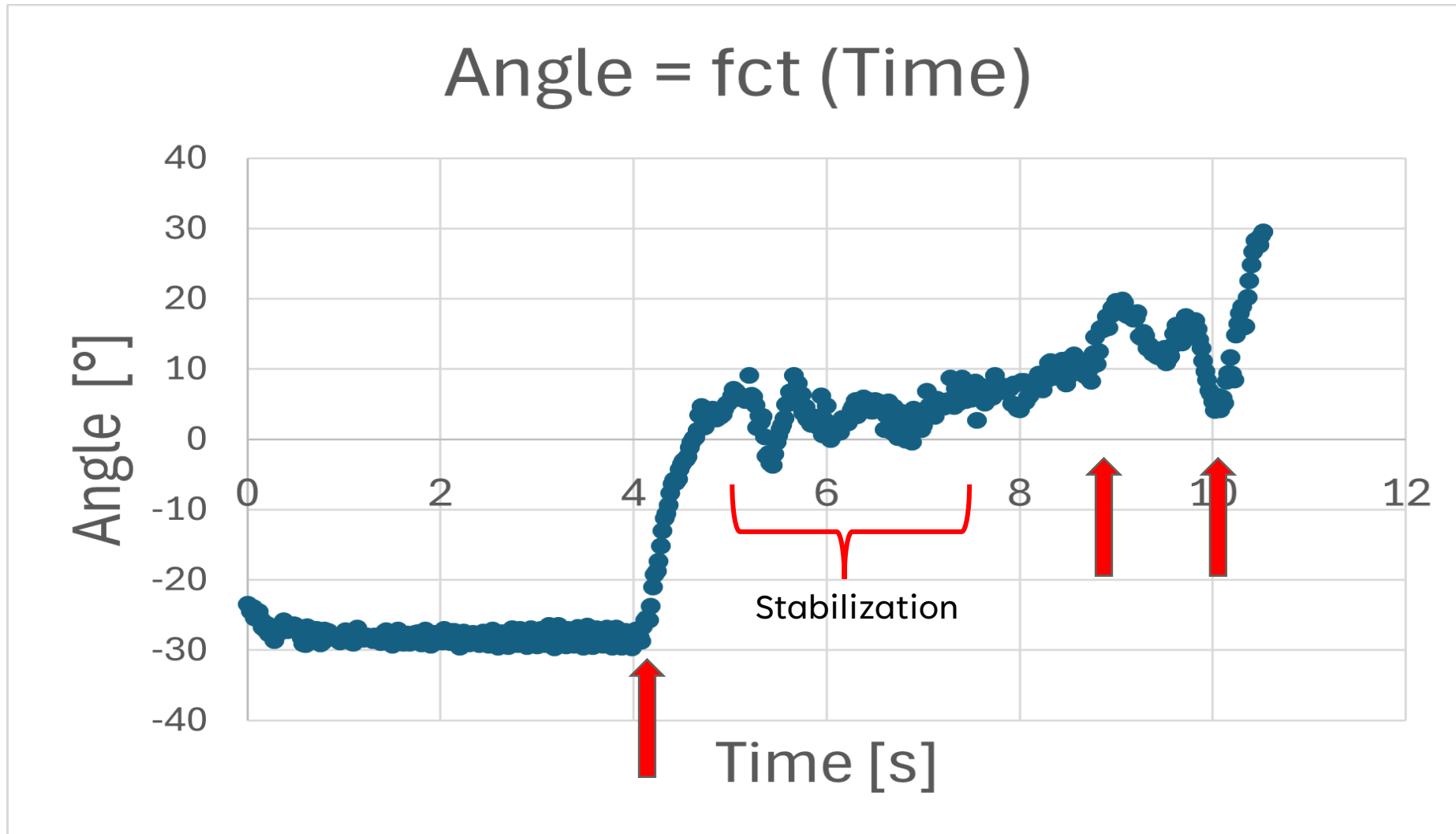
- Start with P only ($K_i = K_d = 0$)
 - Increase K_p until the response becomes fast with a small overshoot.
 - If the response is too slow $\rightarrow$ increase K_p .
- Add integral action
 - Increase K_i to remove steady-state error.
 - If oscillations grow or the system becomes unstable $\rightarrow$ decrease K_i .
- Add derivative action
 - Increase K_d to damp oscillations and reduce overshoot.
 - If the control becomes noisy $\rightarrow$ decrease K_d .



ACCOMPLISHMENT

- Stabilize with the iron bearings
- Regulate the speed with the second PID

CURVES AND GRAPHS



LIMITATIONS AND FUTURE WORK

- Structural conflict between angle and speed loops
 - > motor shacking
- Energy-based swing-up control

An abstract graphic consisting of two thin, dark grey lines intersecting on a light grey background. One line is oriented diagonally from the top-left towards the bottom-right, while the other is oriented from the top-right towards the bottom-left. The intersection point is located in the upper-left quadrant of the image.

TEAM REFLECTION